

1.1 Introduction to Systems of Linear Equations

CHAPTER 1 OVERVIEW – Systems of Linear Equations and Matrices

Information organized into rows and columns forms a rectangular array called a **matrix**. For example, the system

$$5x + y = 3, \quad 2x - y = 4$$

is fully captured by the matrix $\begin{bmatrix} 5 & 1 & 3 \\ 2 & -1 & 4 \end{bmatrix}$, and solving the system amounts to performing operations on this matrix. The study of matrices and related topics is called **linear algebra** – the subject of this chapter.

1.1 Introduction to Systems of Linear Equations

Linear systems and their solutions are a central topic of this course. This section introduces the basic terminology and a method for solving them.

Linear Equations

Definition & Key Rules

Definition (Linear Equation). A linear equation in $x_1, x_2, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b \tag{1}$$

with constants $a_1, \dots, a_n, b$, not all $a_i = 0$. For $n = 2$ or 3 we often drop the subscripts:

$$a_1x + a_2y = b, \quad a_1x + a_2y + a_3z = b$$

When $b = 0$, Equation (1) is called a **homogeneous linear equation**.

Definition (System and Solution). A finite collection of linear equations is a **system of linear equations** (or **linear system**); its variables are **unknowns**. A **solution** is a sequence $s_1, \dots, s_n$ that makes every equation true when substituted for $x_1, \dots, x_n$, written as the **ordered n -tuple**

$$(s_1, s_2, \dots, s_n)$$

– an **ordered pair** if $n = 2$, an **ordered triple** if $n = 3$.

EXAMPLE 1 | Linear Equations

A linear equation contains no products or roots of variables – every variable appears only to the first power, never inside a trigonometric, logarithmic, or exponential function. Linear:

$$x + 3y = 7, \quad x_1 - 2x_2 - 3x_3 + x_4 = 0, \quad \frac{1}{2}x - y + 3z = -1, \quad x_1 + x_2 + \cdots + x_n = 1$$

Not linear:

$$x + 3y^2 = 4, \quad 3x + 2y - xy = 5, \quad \sin x + y = 0, \quad \sqrt{x_1} + 2x_2 + x_3 = 1$$

Linear Systems in Two and Three Unknowns

A linear system in two unknowns,

$$a_1x + b_1y = c_1, \quad a_2x + b_2y = c_2,$$

has two lines as its graph; each solution (x, y) is a point where the lines meet, so three cases arise (**Figure 1.1.1**):

1. Parallel, distinct lines: no intersection, so no solution.
2. Lines meeting at one point: exactly one solution.
3. Coincident lines: infinitely many intersection points, hence infinitely many solutions.

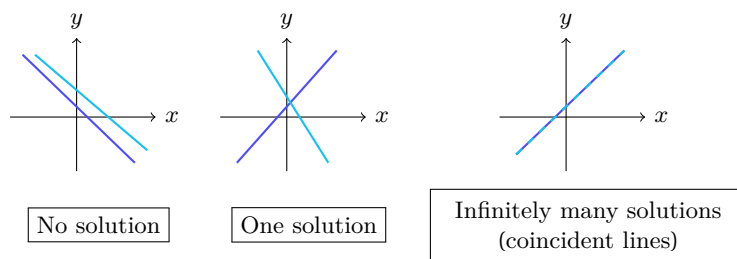


FIGURE 1.1.1

A linear system is **consistent** if it has at least one solution, **inconsistent** if it has none. A consistent system of two equations in two unknowns thus has one solution or infinitely many – no other option. The same holds for three equations in three unknowns,

$$a_1x + b_1y + c_1z = d_1, \quad a_2x + b_2y + c_2z = d_2, \quad a_3x + b_3y + c_3z = d_3,$$

whose graphs are planes: the solution set, if nonempty, is where all three planes meet, again giving no solution, one solution, or infinitely many (**Figure 1.1.2**).

These observations hold for *all* linear systems (proved later):

Theorem

Every system of linear equations has **zero, one, or infinitely many solutions**. There are no other possibilities.

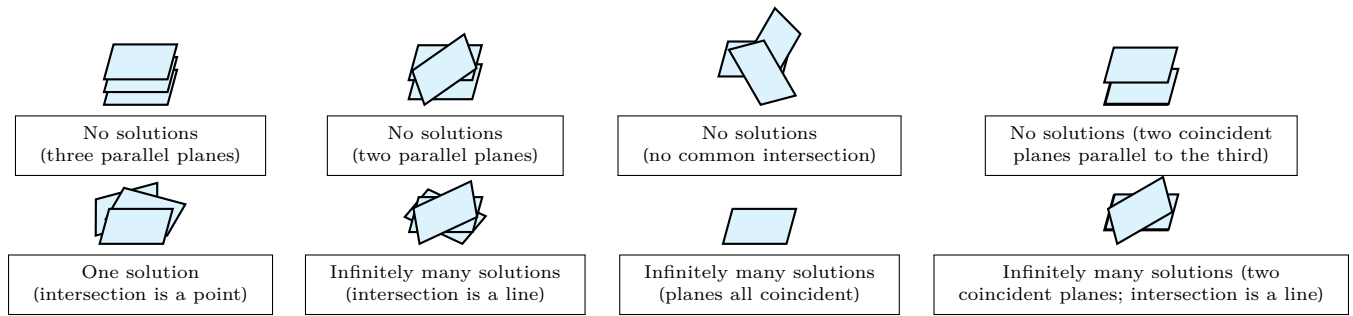


FIGURE 1.1.2

EXAMPLE 2 | A Linear System with One Solution

Solve the linear system

$$x - y = 1, \quad 2x + y = 6$$

SOLUTION: Add -2 times equation 1 to equation 2 to eliminate x :

$$x - y = 1, \quad 3y = 4$$

Then $y = \frac{4}{3}$, and substituting back gives $x = 1 + y = \frac{7}{3}$. Unique solution:

$$x = \frac{7}{3}, \quad y = \frac{4}{3}$$

Geometrically, the lines meet at the single point $(\frac{7}{3}, \frac{4}{3})$.

EXAMPLE 3 | A Linear System with No Solutions

Solve the linear system

$$x + y = 4, \quad 3x + 3y = 6$$

SOLUTION: Add -3 times equation 1 to equation 2:

$$x + y = 4, \quad 0 = -6$$

The contradiction $0 = -6$ means the system has no solution – geometrically, the lines are parallel and distinct.

EXAMPLE 4 | A Linear System with Infinitely Many Solutions

Solve the linear system

$$4x - 2y = 1, \quad 16x - 8y = 4$$

SOLUTION: Add -4 times equation 1 to equation 2:

$$4x - 2y = 1, \quad 0 = 0$$

Since $0 = 0$ imposes no restriction, it can be dropped, leaving the single equation

$$4x - 2y = 1 \tag{8}$$

so the two original lines coincide. Solving (8) for x gives $x = \frac{1}{4} + \frac{1}{2}y$; letting $y = t$ (a **parameter**) yields the **parametric equations**

$$x = \frac{1}{4} + \frac{1}{2}t, \quad y = t$$

For example, $t = 0, 1, -1$ give the solutions $(\frac{1}{4}, 0), (\frac{3}{4}, 1), (-\frac{1}{4}, -1)$.

EXAMPLE 5 | A Linear System with Infinitely Many Solutions

Solve the linear system

$$x - y + 2z = 5, \quad 2x - 2y + 4z = 10, \quad 3x - 3y + 6z = 15$$

SOLUTION: Equations 2 and 3 are multiples of equation 1, so the three planes coincide; it suffices to solve

$$x - y + 2z = 5 \tag{9}$$

Solving for x and setting $y = r, z = s$ (parameters) gives the parametric equations

$$x = 5 + r - 2s, \quad y = r, \quad z = s$$

For example, $r = 1, s = 0$ gives the solution $(6, 1, 0)$.

Augmented Matrices and Elementary Row Operations

Definition & Key Rules

Definition (Augmented Matrix). For the linear system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

is its **augmented matrix**.

Key Rule (Elementary Row Operations). Solving a linear system means applying operations that preserve its solution set:

1. Multiply an equation (row) through by a nonzero constant.
2. Interchange two equations (rows).
3. Add a constant times one equation (row) to another.

Performed on the rows of an augmented matrix, these are the **elementary row operations**.

EXAMPLE 6 | Using Elementary Row Operations

Left: solving by operating directly on the equations. Right: the equivalent row operations on the augmented matrix.

Initial linear system	
$x + y + 2z = 9$ $2x + 4y - 3z = 1$ $3x + 6y - 5z = 0$	$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$
Add -2 times the first equation/row to the second to obtain	
$x + y + 2z = 9$ $2y - 7z = -17$ $3x + 6y - 5z = 0$	$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 3 & 6 & -5 & 0 \end{bmatrix}$
Add -3 times the first equation/row to the third to obtain	

$x + y + 2z = 9$ $2y - 7z = -17$ $3y - 11z = -27$	$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{bmatrix}$
Multiply the second equation/row by $\frac{1}{2}$ to obtain	
$x + y + 2z = 9$ $y - \frac{7}{2}z = -\frac{17}{2}$ $3y - 11z = -27$	$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{bmatrix}$
Add -3 times the second equation/row to the third to obtain	
$x + y + 2z = 9$ $y - \frac{7}{2}z = -\frac{17}{2}$ $-\frac{1}{2}z = -\frac{3}{2}$	$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{bmatrix}$
Multiply the third equation/row by -2 to obtain	
$x + y + 2z = 9$ $y - \frac{7}{2}z = -\frac{17}{2}$ $z = 3$	$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$
Add -1 times the second equation/row to the first to obtain	
$x + \frac{11}{2}z = \frac{35}{2}$ $y - \frac{7}{2}z = -\frac{17}{2}$ $z = 3$	$\begin{bmatrix} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$
Add $-\frac{11}{2}$ times the third equation/row to the first and $\frac{7}{2}$ times the third to the second to obtain	
$x = 1$ $y = 2$ $z = 3$	$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$

The solution $x = 1$, $y = 2$, $z = 3$ is now evident.

Exercise Set 1.1

- Determine whether each equation is linear in x_1, x_2, x_3 .
(a) $x_1 + 5x_2 - \sqrt{2}x_3 = 1$ (b) $x_1 + 3x_2 + x_1x_3 = 2$ (c) $x_1 = -7x_2 + 3x_3$
(d) $x_1^{-2} + x_2 + 8x_3 = 5$ (e) $x_1^{3/5} - 2x_2 + x_3 = 4$ (f) $\pi x_1 - \sqrt{2}x_2 = 7^{1/3}$
- Using the notation of Formula (7), write a general linear system with
(a) two equations in two unknowns. (b) three equations in three unknowns. (c) two equations in four unknowns.
- Find a system of linear equations in $x_1, x_2, x_3, \dots$ corresponding to each augmented matrix.
(a) $\begin{bmatrix} 2 & 0 & 0 \\ 3 & -4 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & 0 & -2 & 5 \\ 7 & 1 & 4 & -3 \\ 0 & -2 & 1 & 7 \end{bmatrix}$
- Find the augmented matrix for each linear system.
(a) $-2x_1 = 6, 3x_1 = 8, 9x_1 = -3$ (b) $6x_1 - x_2 + 3x_3 = 4, 5x_2 - x_3 = 1$
(c) $2x_2 - 3x_4 + x_5 = 0, -3x_1 - x_2 + x_3 = -1, 6x_1 + 2x_2 - x_3 + 2x_4 - 3x_5 = 6$
- Determine whether each 3-tuple is a solution of
$$2x_1 - 4x_2 - x_3 = 1, \quad x_1 - 3x_2 + x_3 = 1, \quad 3x_1 - 5x_2 - 3x_3 = 1$$

(a) $(3, 1, 1)$ (b) $(3, -1, 1)$ (c) $(13, 5, 2)$ (d) $(\frac{13}{2}, \frac{5}{2}, 2)$ (e) $(17, 7, 5)$
- Solve each system, if possible, and state whether the corresponding lines have zero, one, or infinitely many intersection points; give coordinates or parametric equations accordingly.
(a) $3x - 2y = 4, 6x - 4y = 9$ (b) $2x - 4y = 1, 4x - 8y = 2$ (c) $x - 2y = 0, x - 4y = 8$
- Use parametric equations to describe the solution set of each linear equation.
(a) $7x - 5y = 3$ (b) $3x_1 - 5x_2 + 4x_3 = 7$ (c) $-8x_1 + 2x_2 - 5x_3 + 6x_4 = 1$ (d) $3v - 8w + 2x - y + 4z = 0$
- Each system below has infinitely many solutions; describe each solution set parametrically.
(a) $2x - 3y = 1, 6x - 9y = 3$ (b) $x_1 + 3x_2 - x_3 = -4, 3x_1 + 9x_2 - 3x_3 = -12, -x_1 - 3x_2 + x_3 = 4$
- Find one elementary row operation that creates a 1 in the upper-left corner of the given augmented matrix without introducing fractions in the first row.
(a) $\begin{bmatrix} -3 & -1 & 2 & 4 \\ 2 & -3 & 3 & 2 \\ 0 & 2 & -3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & -1 & -5 & 0 \\ 2 & -9 & 3 & 2 \\ 1 & 4 & -3 & 3 \end{bmatrix}$
- Find all values of k for which the given augmented matrix corresponds to a consistent linear system.
(a) $\begin{bmatrix} 1 & k & -4 \\ 4 & 8 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & k & -1 \\ 4 & 8 & -4 \end{bmatrix}$
- The curve $y = ax^2 + bx + c$ shown passes through $(x_1, y_1), (x_2, y_2), (x_3, y_3)$. Show that a, b, c solve the system with augmented matrix

$$\begin{bmatrix} x_1^2 & x_1 & 1 & y_1 \\ x_2^2 & x_2 & 1 & y_2 \\ x_3^2 & x_3 & 1 & y_3 \end{bmatrix}$$

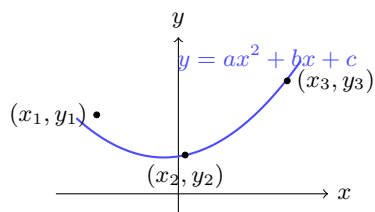


FIGURE Ex-21

22. Explain why none of the three elementary row operations changes the solution set of a linear system.
23. Show that if $x_1 + kx_2 = c$ and $x_1 + lx_2 = d$ have the same solution set, then $k = l$ and $c = d$.
25. A diet requires 7 units of fat, 9 of protein, and 16 of carbohydrates per meal, drawn from three foods (per ounce):
 - Food 1: 2 fat, 2 protein, 4 carbohydrates.
 - Food 2: 3 fat, 1 protein, 2 carbohydrates.
 - Food 3: 1 fat, 3 protein, 5 carbohydrates.
 With x, y, z the ounces of foods 1, 2, 3, find (but do not solve) the linear system whose solution gives the required amounts.